Joshua Sun

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Education

San José State University — San José, CA
B.S. in Applied Mathematics, Minor in Computer Science — *Expected graduation: Fall 2027*

Projects

Unitern | *TypeScript, Next.js, PostgreSQL, GCP* — Independent Project

- Built a **Next.js/React/PostgreSQL** housing marketplace with listing management, messaging, map search, Google OAuth, single-use account tokens, and atomic SQL rate limits.
- Added **unit, integration, and Playwright tests**; built Docker/GitHub Actions releases and Terraform for Cloud Run, IAM, and Secret Manager, with pre-deploy migrations and smoke checks.

GNN Feasibility Auditor | *Python, FastAPI, React, Docker* — Independent Project

- Built an **exact 1-WL graph auditor** with independent certificate replay, a FastAPI job API, React/Cytoscape visualizations, an installed CLI, and offline HTML reports.
- Implemented isolated workers, bounded queues, cancellation, and time/memory limits; checked **5,265 graph pairs** against an oracle and recorded a **2.31 s local median** on a **50,000-node path** across three probes.

Beacon | *TypeScript, React, MediaPipe* — LA Hacks — Team Project

- **3rd Place, Best Design — Paper Sponsor Track (LA Hacks team award).**
- Built browser-based gesture recognition using **32-dimensional embeddings** and a personalized **5-nearest-neighbor classifier**, calibrating from **15 examples per class** for caregiver requests.
- Implemented sustained-gesture confirmation, camera-stall resets, and an **8-second cooldown**; connected CREAO adaptation to caregiver-approved, bounded configuration updates.

Parkii | *Python, TypeScript, React, Convex, YOLO* — Cal Hacks — Team Project

- Co-developed a camera-based parking assistant connecting **YOLO11s/OpenCV**, Python feed workers, Convex real-time state, calibrated stall geometry, and a React map.
- Integrated a **Claude tool-use voice agent** for camera inspection, spot selection, app-level holds, and navigation, with Cartesia speech streamed over **WebSocket**.

Lox Language Interpreter | *Go, CLI, Language Implementation* — In Progress

- Developing a Go implementation of *Crafting Interpreters*; built file execution and interactive input scaffolding, error reporting, exit-code handling, and scanner integration points.

Technical Skills

Languages: Python, Java, C++, Go, TypeScript/JavaScript, SQL
Frameworks: React, Next.js, Node.js, Express.js, FastAPI, Spring Boot
Databases & Messaging: PostgreSQL, MySQL, Redis, Apache Kafka
Cloud & DevOps: AWS, Docker, Kubernetes, Terraform, GitHub Actions
Testing & Tools: Git, Linux, pytest, JUnit, Playwright
AI Development: Codex, Claude Code, LangChain, LangGraph, RAG, MCP

Certifications

AWS Certified Solutions Architect – Associate — *In progress*

AWS Certified Machine Learning Engineer – Associate — *In progress*